

IAU_{THESIS}
DOCUMENTATION

Documentation and Example of the
IAU thesis L^AT_EX Documentclass

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I want to thank all students of the IAU. Without you this would have been buried by time. You kept asking for a template and now this thing exists. I hope this is of any use to you.

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About this Documentclass

The `IAUthesis` is an unofficial documentclass for writing a thesis or a project report at the Institute for Atmospheric and Environmental Sciences at the Goethe University Frankfurt. This document you are currently reading functions both as a deocumentation and an example of the IAUthesis documentclass.

This started with my master thesis, for which I put a lot of work into exactly recreating the layout in L^AT_EX which the examination office was asking for. Later I made some changes to this layout for my master thesis. Now this has finally evolved into a documentclass, which makes it much nicer to handle and provides you with many options.

If you find any bug or want features, please reach out to me. You find my contact information on the bottom left of the titlepage. Ideally you go to github ([link](#)), create a fork, and then later make a pull request with the bugfix or feature. You can also create an issue ([link](#)) there.

I wish you good luck and all the best for your thesis or report!

Best, Frederik

Documentclass Options

The Documentclass options are what the “words” you put into the rectangular brackets of the `\documentclass[<options>]{IAUthesis}` command, which usually sits in the first line of the document and tells L^AT_EX a lot about the layout. The options activate or deactivate features, can change the layout, and other settings.

2.1 The minimum you need

If you just want to write you don’t need to pass any class options although you might want to use `11pt` or `12pt` to adjust the documents fontsize.

I’d also recommend to use `todonotes` as option. This enables you write notes wherever you want in the text simply by writing `\todo{ToDo: Nothing.}`. When you want to hand your thesis in, just remove this option and all notes in the PDF will disappear.

ToDo:
Nothing.

For the *project report*, which typically isn’t printed, you might want to consider the `oneside` and `nodeclaration` options.

Finally you may want to change the language of the document to German. In this case you simply need to pass the option `german` to the documentclass.

In conclusion the recommended documentclass command should look similar to one of the following:

```
\documentclass[11pt, todonotes]{IAUthesis}
```

```
\documentclass[11pt, todonotes, german]{IAUthesis}
```

```
\documentclass[12pt]{IAUthesis}
```

```
\documentclass[12pt, oneside, nodeclaration]{IAUthesis}
```

However, there are more options, which you can discover in the following section.

2.2 Description of all Documentclass Options

german Changes the document language to German, affecting titles and default texts.

english (default) Provides English titles and texts.

noabstract Hides the abstract section.

noacknowledgements Hides the acknowledgements section.

nodeclaration Hides the declaration page (CAUTION! important for a thesis).

notoc Removes the table of contents from the frontmatter.

notof Removes the list of figures from the frontmatter.

notot Removes the list of tables from the frontmatter.

noams Prevents the default loading of `amsmath`, `amsthm` and `amssymb`.

oneside Uses a single-sided layout suitable for the project report or PDF distribution.

twoside (default) This option provides a two-sided layout for print. Especially for an adhesive binding, which is typical for a Bachelor or Master thesis.

fancypagenumbers Changes the page-numbering in the frontmatter to lowercase roman numbers and counts them separately.

largemargins Increases the margins around the text. Reduces the text-area from $23\text{cm} \times 16\text{cm}$ to $20\text{cm} \times 15\text{cm}$, which can be favourable for a Bachelor Thesis.

todonotes Enables todonotes: This enables you to write notes wherever you want in the text by writing `\todo{HI!}`.

HI!

You can change the colors of the notes: `\todo[color=blue!30!white]{lightblue}`.

lightblue

With `\listoftodos` you get a list of all your notes. You find an example at the end of the document. When you want to hand your thesis in, just remove the option `todonotes` from the `\documentclass[]{\IAUthesis}` command and all notes in the PDF will disappear.

Customizing the Document

There are several commands to customize the document further after you have chosen the documentclass options you like. Everything here goes into the “head” of the document, this means it goes after `\documentclass[]{}{}` and before `\begin{document}`.

`\title{<text>}` The title of your document. Example: `\title{Topic of my Thesis}`

`\author{<name>}` The author name(s) printed on the title page.

Example: `\author{My Name}`

`\date{<date>}` The date printed on the title page. Example: `\date{19.02.2026}`

`\firstreviewer{<name>}` Sets the first reviewer name used in the Supervisor block.

Example: `\firstreviewer{Prof. Lorem}`

`\secondreviewer{<name>}` Sets the second reviewer name used in the contact block.

Example: `\secondreviewer{Dr. Ipsum}`

`\abstracttext{<text>}` Sets the text for the abstract section in the frontmatter. You can write a lot and also use linebreaks. Example:

`\abstracttext{This thesis investigates...}`

`\acknowledgements{<text>}` Sets the text for the acknowledgements section in the frontmatter. You can write a lot and also use linebreaks.

Example: `\acknowledgements{Thanks to colleagues and family.}`

`\thesistype{<text>}` Sets the thesis type label shown on the title page (e.g., “Thesis”, “Master Thesis”). Example: `\thesistype{Master Thesis}`

`\thesisdegree{<text>}` Sets the degree line printed under the thesis type (e.g., “MSc in Something”). Example:

```
\thesisdegree{M.Sc. Atmospheric and Climate Sciences}
```

`\affiliation{<text>}` (optional) Sets the affiliation block shown on the title page. The *Institute for Atmospheric and Environmental Sciences* is set as a default.

Example: `\affiliation{Institute,\\University}`

`\declarationtext{<text>}` (optional) Changes the Text of the declaration. You can get the default text with the command `\declarationdefault`. Example:

```
\declarationtext{\declarationdefault\bigskip\par
\noindent This is an additional sentence in the declaration.}
```

`\contact{<text>}` (optional) CAUTION! Overrides the supervisors block. When this command is used no supervisors will be displayed, only the contact info. Example:

```
\contact{Contact:\\contact@frederik.vieirafischer.de\\@flvf:matrix.org}
```

`\newabbreviation{<tag>}{<abbreviation>}{<full-text>}` Define a new abbreviation which can be used in the document and will be explained, when it is first called and in the Abbreviations section in the appendix.

Example: `textbackslash newabbreviation{bsc}{B.Sc.}{Bachelor of Science}`

For further information refer to the manual:

<https://ctan.org/pkg/glossaries-extra>

`\glssetwidest{<abbreviation>}` Sets the length of the longest abbreviation for the *Abbreviations* Section. Example: `\glssetwidest{B.Sc.}`

The Body of the Document

This chapter describes the body of the documents as well as some options provided to you. This is everything between `\begin{document}` and `\end{document}`.

4.1 Document Structure

The document usually consists of the frontmatter, the main content, and the appendix. Each of the sections are explained shortly in the following. Throughout the entire document the text-area has the size of $23\text{ cm} \times 16\text{ cm}$.

4.1.1 Frontmatter

This starts with the titlepage and includes the declaration, acknowledgements, abstract, table of contents, list of figures, and list of tables. This documentclass is setup in a manner, that you don't need to worry about any of that. The `\maketitle` command takes care of it, however you need to provide some information. If you want to remove a section from the frontmatter, refer to the documentclass options. Example:

```
\documentclass[11pt]{IAUthesis}
\title{<title>}
\author{<author>}
\date{<hand-in date>}
\firstreviewer{<first reviewer name>}
\secondreviewer{<second reviewer name>}
\thesistype{e.g. Master Thesis}
\thesisdegree{e.g. B.Sc. Meteorology}
```

```
\acknowledgements{<acknowledgements>}
\abstracttext{<abstracttext>}
\begin{document}
\maketitle
\chapter{<first chapter title>}
```

4.1.2 Main Content

Here is where all your text, formulas, figures, and so fourth reside. You typically split it into major parts like the introduction, the objectives, the theoretical background, the methodology, the results, and the discussion. This can look like this:

```
\chapter{Methodology}
These are interesting methods.
\section{First Method}
I did this.
\subsection{A Detail}
This is part of the first method.
\subsection{Another Detail}
Also belongs to the first method.
\section{Second Method}
I also did this.
```

In theory you can also use `\subsubsection` and `\subsubsubsection`, however, I wouldn't recommend that, because from the perspective of the reader the document will lack internal cohesion. Ask yourself: "Is this really another section or just another paragraph?"

You can also use numbered and unnumbered lists, as well as starred (sub)sections (`\section*{}`). The latter will appear as a heading, but without a number and without appearing in the table of contents. This can be a good tool to add a bit of structure without emphasising the section as being a "stand-alone" thing of itself.

4.1.3 Appendix

This is at the end of the document, where the abbreviations, additional material, and the bibliography reside. You can add whatever you want to the additional materials and create as many chapters as you like. They work the same as chapters in the main content, but are marked as appendix. Example:

```
\chapter{Discussion}
```

```
This is the last part of my main content.
```

```
Hereafter, only the appendix follows.
```

```
\cleardoublepage\appendix\pagestyle{appendix}
```

```
\printnoidxglossary[title={\abbrtitle}]
```

```
\chapter{Additional Material}
```

```
Here I can add as many figures, tables, and as much code as I want.
```

```
Anything my heart desires.
```

```
\printbibliography[heading=bibnumbered, title={\bibtitle}]
```

```
\end{document}
```

4.2 Useful Commands and Tricks

In this section I'll present you a few commands and useful tricks this documentclass is set up to perform.

4.2.1 Paragraphs and Pagebreaks

The first and most important thing: **Do not use the double backslash!** (No `\\` !)
They are very inflexible and will make L^AT_EX struggle trying to set your document. If you want to start a new paragraph, you just need to leave an empty line before continuing with your text. If you want to have more space between two paragraphs, there are three

commands you can use:

```
1 This is the first paragraph.  
2  
3 This is the second paragraph, directly following the first.\smallskip  
4  
5 This paragraph has a small distance to the previous. \medskip  
6  
7 This one has a medium distance.\bigskip  
8  
9 This paragraph has a big distance to the previous. Isn't that lovely?
```

If you don't want a paragraph starting with an indent, put `\noindent` at the beginning of the line. For page breaks there are two commands.

`\clearpage` This will leave the rest of the page empty and start a new one.

`\cleardoublepage` (twoside documents only) This will leave the rest of the page empty, and if necessary also the next page. Whatever you write hereafter, appears on the next right (odd) page.

4.2.2 Citing Sources and Bibliography

This documentclass comes fully set up for handling sources. You just add your `.bib` library file with `\addbibresource{<.bib file>}` in the head of the document (after `\documentclayy[]{}` and before `\begin{document}`). Your primary two commands to cite sources are `\citep` and `\citete`.

`\citep` This command will put your sources in parentheses:

`\citep{IAUthesis}` will return: (Vieira Fischer, 2026)

`\citep{IAUthesis, IAUthesis}` will return: (Vieira Fischer, 2026; Vieira Fischer, 2026)

`\citet` Is for direct references within a sentence. "`\citet{IAUthesis}` did something." will return: "Vieira Fischer (2026) did something."

The Bibliography is usually at the end of the document as the last chapter in the appendix:

```
\cleardoublepage\appendix\pagestyle{appendix}
...
\printbibliography[heading=bibnumbered, title={\bibtitle}]
\end{document}
```

4.2.3 Figures, Tables, and Captions

For figures and tables you should use the `figure` and `table` environments. For the captions this documentclass further includes the `captions` package, which enable you to write an additional short caption, which will be displayed in the list of figures or list of tables respectively. The `<short caption>` should only hint towards the contents of the figure/table and you should not be so long that it leads to a linebreak (although this is sometimes okay). The `<long description>` should describe the figure/table in much more detail. Examples:

```
\begin{figure}[<positioning>]
\centering
\includegraphics[width=<width>]{<file>}
\caption[<short caption>]{<long caption>\label{<label>}}
\end{figure}

\begin{table}[<positioning>]
\centering
\caption[<short caption>]{<long caption>\label{<label>}}
\begin{tabular}{r|r}
<col1> & <col2> \\
\hline
<num1> & <num2> \\
\end{tabular}
\end{table}
```

`\end{table}`

The `<positioning>` uses the following letters to give the interpreter hints where to put the float (figure/table environment):

- h** Place the float approximately here (at the same point it occurs in the source). $\text{\LaTeX}$ will try to honor this but may move the float if it breaks page layout or float placement constraints.
- t** Place the float at the top of a page. Preferred for floats that should appear near the beginning of a page.
- b** Place the float at the bottom of a page. Useful for floats that should follow surrounding text on the same page.
- p** Place the float on a dedicated float page (a page containing only floats). $\text{\LaTeX}$ collects floats and places them together when this is used.
- !** Override internal parameters $\text{\LaTeX}$ uses for float placement (relax constraints). This increases the chance $\text{\LaTeX}$ will place the float according to the other specified options.
- ht** Try here first; if not possible, try top of page.
- hb** Try here first; if not possible, try bottom of page.
- tp** Try top of page first; if not possible, put on float page.
- bp** Try bottom of page first; if not possible, put on float page.
- htb** Try here first; if not possible, try top of page; if that also fails, try bottom of page. This is a common default combination balancing proximity and good page layout.

The `<file>` for the figure can in theory be a .jpg or .png, but a .pdf is recommended. Using `\input{}` a .pgf file is also an option, however this needs more expertise in creation, but will lead to beautiful results. The *width* of the file should not exceed 16 cm (or 15 cm if the document option `largemargins` is used). Ideally The plot itself is created

so it doesn't exceed the page layout and no `width` or `scale` option has to be passed to `\includegraphics[<options>]{<file>}`. If a `<width>` has to be given, some of the following options would work: `10cm`, `\textwidth`, `0.8\linewidth`, etc.

The `<label>` can be almost any string. I like to use `fig:something` for figures, `tab:something` for tables and `sec:something` for sections. With `\ref{<label>}` you can refer to a label within the text.

4.2.4 Abbreviations

This documentclass has the package `glossaries-extra` setup in a way, to handle abbreviations. Before you start using abbreviations, think about which abbreviations are really necessary. Too many abbreviations will make your text difficult to read, however some important concepts are easier to recognise when abbreviated.

Setup

Before you use an abbreviation, you need to set it up in the head of the document (before `\begin{document}`). You can do this with one simple command for each abbreviation:

`\newabbreviation{<tag>}{<abbreviation>}{<full-text>}` You have to give each abbreviation a “tag” which is an unique label by which you can call it. Then the abbreviation is the short form (e.g. B.Sc.) and the full text is the non-abbreviated word or term (e.g. Bachelor of Science). You can add some optional arguments, which you find in the manual: <https://ctan.org/pkg/glossaries-extra>

Example:

```
\documentclass[11pt]{IAUthesis}
\newabbreviation{bsc}{B.Sc.}{Bachelor of Science}
\glssetwidest{B.Sc.}
\begin{document}
\maketitle\glsresetall
\chapter{First Part}
```

Using Abbreviations

The package will keep track of your abbreviations and write out the entire term with the abbreviation in parenthesis, the first time you use it. You use an abbreviation with `\gls{<tag>}`, e.g. `\gls{bsc}` which will produce: Bachelor of Science (B.Sc.) However, any usage thereafter will produce: B.Sc.

There are some important commands to handle your abbreviations:

`\gls{<tag>}` This will insert the abbreviation into the text where this command is used.

`\glspl{<tag>}` Same as `\gls{<tag>}` but plural form.

`\Gls{<tag>}` Same as `\gls{<tag>}` but ensures a capital letter at the start.

`\Glspl{<tag>}` Same as `\glspl{<tag>}` but ensures a capital letter at the start.

`\glsreset{<tag>}` Resets the abbreviation, so that its next usage will appear as the first again.

`\glsresetall` Same as `\glsreset{<tag>}` but for all abbreviations.

`\glssetwidest{<abbreviation>}` This will set the maximum abbreviation width for the list of abbreviations. Example: `\glssetwidest{B.Sc.}`

List of Abbreviations

The list of abbreviations is typically the first thing in the appendix. It can be displayed by using the command `\printnoidxglossary[title={\abbrtitle}]`. Example:

```
\cleardoublepage\appendix\pagestyle{appendix}
\printnoidxglossary[title={\abbrtitle}]
```

4.2.5 Numbers, Values, and Units

To handle units properly this documentclass makes use of the package `siunitx`. This package primarily provides the following commands:

`\SI{<number>}{<unit>}` This command should be used everywhere when you give a number with a unit. The number can also include an error. Examples:

`\SI{9.81}{\meter\per\square\second}` returns: $9,81\text{ ms}^{-2}$

`\SI{54.3+-2.1}{\degree N}` returns: $(5,43 \pm 0,21)^\circ\text{N}$

`\SIRange{<start>}{<end>}{<unit>}` This will create a range with a unit:

`\SIRange{5e3}{7er}{\kelvin}` returns: $5 \cdot 10^3 - 7 \cdot 10^3\text{ K}$

`\si{<unit>}` Will only display a unit: `\si{\celsius}` returns: $^\circ\text{C}$

`\num{<number>}` Will display a number:

`\num{2.7315e2}` returns: $2,7315 \cdot 10^2$

`\num{3.141+- .008}` returns: $3,14 \pm 0,01$

For more information refer to the manual: <https://ctan.org/pkg/siunitx>

4.2.6 Quotes

This documentclass uses the `csquotes` package by default and will automatically change "text" to “text”. For further commands and information refer to the manual: <https://ctan.org/pkg/csquotes>

4.2.7 Notes and Todos (todonotes)

Sometimes you are in the flow of writing and want to make a note, to remind you to look up a source. The package `todonotes` makes this quite easy. Please refer to the documentclass options for further information (Section 2.2).

Appendix A

Abbreviations

B.Sc. Bachelor of Science

Appendix B

Document Template

```
\documentclass[11pt, todonotes]{IAUthesis}

% document informations
\title{<title>}
\author{<author>}
\date{<hand-in date>}
\firstreviewer{<first reviewer name>}
\secondreviewer{<second reviewer name>}
\thesistype{e.g. Master Thesis}
\thesisdegree{e.g. B.Sc. Meteorology}
\acknowledgements{<acknowledgements>}
\abstracttext{<abstracttext>}

% references
\addbibresource{library.bib}

% colors
\definecolor{hotpink}{rgb}{1.00, 0.00, 0.83}

% abbreviations
\newabbreviation{bsc}{B.Sc.}{Bachelor of Science}
\glsssetwidest{B.Sc.~~} % The tildes produce some extra space.
```

```
\begin{document}
\maketitle\glsresetall
\chapter{<first chapter title>}
This is my introduction.
This template was created by \citet{IAUthesis}.
This is the "first paragraph".

This is the second paragraph, directly following the first \smallskip

This paragraph has a small distance to the previous. \medskip

This one has a medium distance.\bigskip

This paragraph has a big distance to the previous.
Isn't that lovely? \todo[color=hotpink]{add source}

\chapter{Methodology}
These are interesting methods.

\section{First Method}
I did this. \Glspl{bsc} is the plural of \gls{bsc} when using abbreviations.

\subsection{A Detail}
This is part of the first method.

\subsection{Another Detail}\glsreset{bsc}
Also belongs to the first method.
Here I have reset the \gls{bsc} abbreviations so it shows long again.
\cleardoublepage
```

`\section{Second Method}`

I also did this `\citep{IAUthesis}`.

Don't refer to Figure `\ref{fig:label}` (it is `\SI{5}{\centi\meter}` wide).

Rather look at Table `\ref{tab:label}`!

`\begin{figure}[htbp]`

`\centering`

`\includegraphics[width=5cm]{GU-Logo-blau-gross.png}`

`\caption[<short caption>]{<long caption>\citep{GUlogo}\label{fig:label}}`

`\end{figure}`

`\begin{table}[htbp]`

`\centering`

`\caption[<short caption>]{<long caption>\label{tab:label}}`

`\begin{tabular}{r|r}`

`<col1> & <col2> \\`

`\hline`

`<num1> & <num2> \\`

`\end{tabular}`

`\end{table}`

`\chapter{Discussion}`

This is the last part of my main content.

Hereafter, only the appendix follows.

`\cleardoublepage\pagestyle{appendix}\appendix`

`\printnoidxglossary[title={\abbrtitle}]`

`\chapter{Additional Material}`

Here I can add as many figures, tables, and as much code as I want.

Anything my heart desires.

```
\printbibliography[heading=bibnumbered, title={\bibtitle}]
```

```
\listoftodos
```

```
\end{document}
```

References

Vieira Fischer, Frederik. L. (2026). *IAUthesis LaTeX documentclass*. Frankfurt. URL:
`github.com/FrederikVF/IAUthesis`.

Todo list

■	ToDo: Nothing.	2
■	HI!	3
■	lightblue	3